

Title:

Ensuring the correct routing of electronic product verification messages for physical Identifiers through the use of domain or subdomain message routing through the manufacturer's public URL

Abstract: In an anti-counterfeit system and method unique product identifiers or codes are used to identify each unique item manufactured allowing each single item to be distinguished from all other alike items. Furthermore, the manufacturers of these items provide an electronic verification capability for each unique code they have used to label an item. This invention simply requires these messages to be routed directly to the public website Uniform Resource locator URL domain or subdomain and then to be redirected to either the manufacturer's own electronic verification capability, or, as is more commonly practised, on to their respective third party service provider, thus providing surety that the verifying party controls both the public facing URL, and any redirects it then makes, of the product being queried for verification.

Background:

The 2014 Drug Supply chain Security Act (the DSCSA) Requires that all Pharmaceutical product be unique labelled and that correctly credentials third parties can verify those labels with the manufacturer during the entirety of a product's lifecycle. As the DSCSA does not stipulate the use of a centralized database, each manufacturer is responsible for verifying their own product. Third party IT suppliers can be used to manage these verifications for manufacturers. The problem occurs when a bad actor steps in and claims to be the authorized manufacturer of a product. Their IT service provider may also be a bad actor, in which case there is currently no way to ascertain at scale if the actual manufacturer is receiving and verifying the verification requests or if an imposter has taken their place. By agreeing to route all messages through the public URL of a manufacturer by utilization of an MX record, the requestor can see that the responder has control of that company's internet presence and therefor decide if they trust that response or not. While this problem has manifested itself in the identification of the correct party for pharmaceutical manufacturer for goods destined for sale in the United States, it is applicable to all unique labelled products that require electronic verification of label issuance.

Drawings:

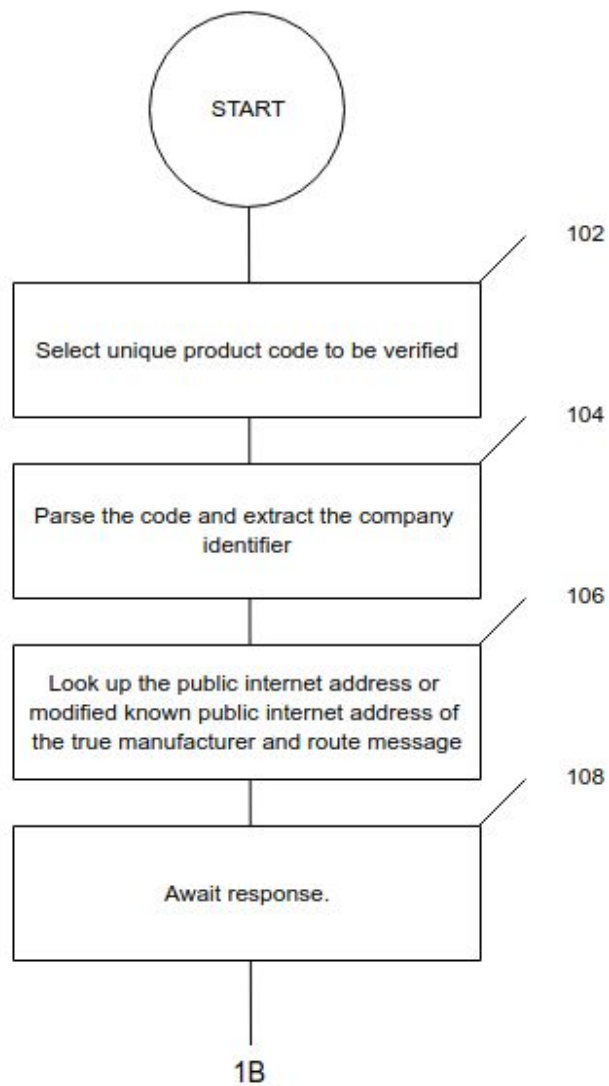


FIGURE 1A

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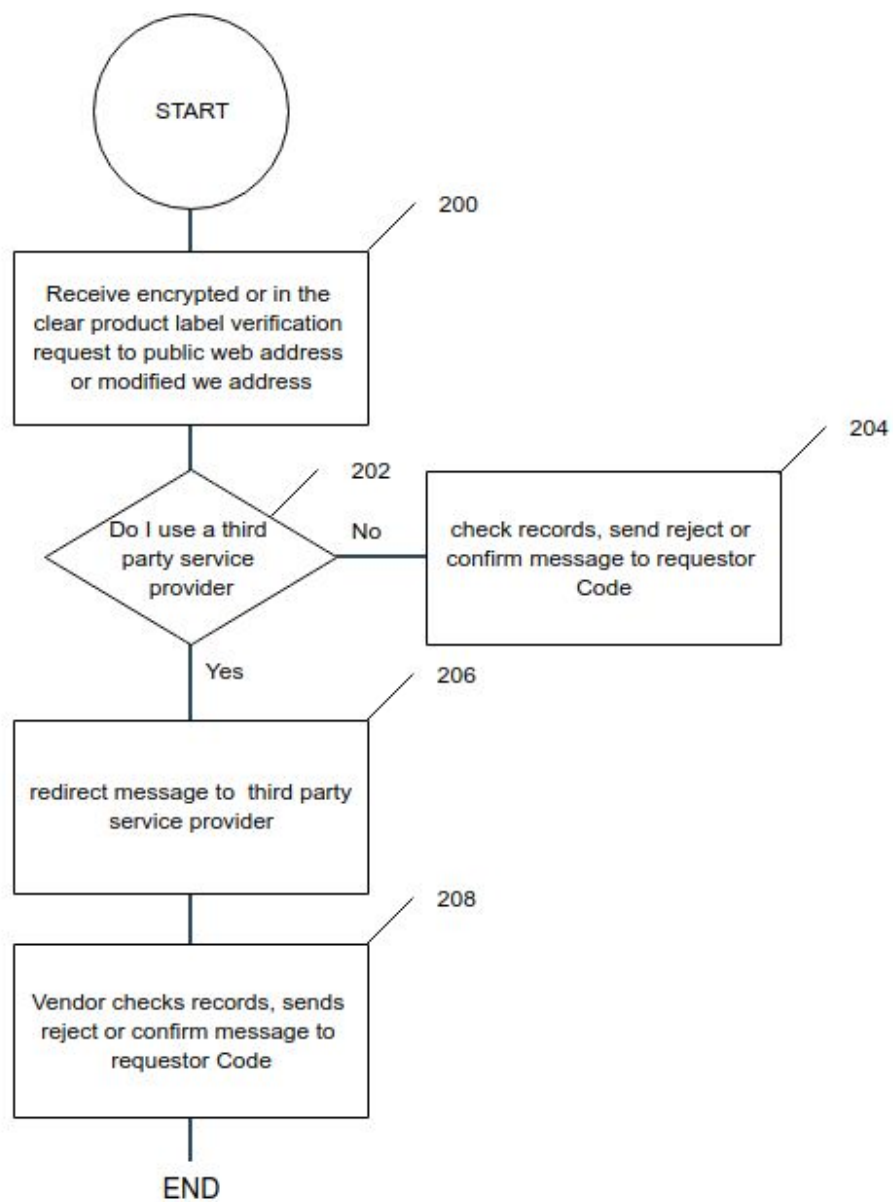


FIGURE 2A

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Background of the invention

While the implementation of standards in supply chain labeling has facilitated the rapid adoption of unique labeling technology and brings benefit to supply chain visibility, it has also offered malicious actors and counterfeiters the capability of understanding the structure of a label and then forward predicting future codes even before manufacturers themselves have issued them. Even to the trained eye these codes look genuine, and so a verification process can be used to check the codes with the original manufacturer to ensure that they were issued, and to record time and location data for every verification so that the manufacturer can flag up unusual behaviour and inform the enquiring party not to use that item. This invention generally relates to this systemic problem of correctly identifying a genuine manufacturer from a fake party, when the rights to electronically confirm a product code is good are claimed. This problem occurs in industries where the verification system is not currently centralized and or not all manufacturers join the system, allowing bad actors to stake claims for products they do not really manufacture and then selling those items as genuine, while also faking the verification responses of the item.

Summary of the Disclosure

Certain embodiments of the present invention allow for the use of a publicly known web address to be required to be utilized in the verification of a products unique identification codes, that web address or sub address is responsible for redirecting any enquiry to the correct verification service provider or system but because the address is publicly known and associated with the product, allows the verifying party to ascertain that the product is being verified by the correct party and not a bad actor.

Brief description of the drawings

Figures. 1A, shows a flow chart of the requesting party for a verification request's side of one embodiment of an anti-counterfeiting system in accordance with the invention.

Figure 2A is a flow chart showing the responding side of one embodiment of an anti-counterfeiting system in accordance with the invention.

Like reference numbers and designations in the various drawings indicate like elements.

A detailed description of the invention

Fig. 1A shows the flow charts for a party initiating an electronic verification request via the public or modified public URL of the genuine manufacturer of the product.

The verifying party reads the code from the labelled item (step 102) and extracts the company identifier from the data contained within the code (step 104).

The verifying party then looks up the manufacturers publicly available website www.genuinemanufacturer.com and routes a verification request to that domain or a subdomain such as verify.genuinemanufacturer.com (step 106).

The verifying party can then await a response, knowing the respondent must have administration control of the public website to be able to either responded directly or to have forwarded the request to their preferred third party verification service provider.

Fig. 2A shows a flow chart for the responding side of one embodiment of an anti-counterfeiting system in accordance with the invention.

A genuine manufacturer with a public URL receives an encrypted or in the clear verification request to their public website address such as www.genuinemanufacturer.com or their modified public website address such as verify.genuinemanufacturer.com (step 200).

If they manage their own verifications (item 202) they unencrypt the request, check the item's product codes against their manufacturing records and respond to the requesting party (item 204).

If they use a third party service provider they forward the message to the vendor who performs the checks and responds on their behalf.

Claims

What is claimed for:

1. An anticounterfeiting method, that allows requesting parties within a network of computers to check electronically that a label was issued by the manufacturer
 - a. By routing messages to the publicly available and accessible website URL or modified website URL of a product manufacturer and receiving a response back directly from the manufacturer, thus giving surety that the responding party is the genuine product manufacturer because the requestor has now checked that they have control over the manufacturers public URL.
 - b. By routing messages to the publicly available and accessible website URL or modified website URL of a product manufacturer and receiving a response back from the manufacturers designated third party verification systems vendor who has been forwarded the message from the manufacturer, thus giving surety that

the responding party is the genuine product manufacturer because the requestor has now checked that they have control over the manufacturers public URL.

ENDS